



Review article

Ambient air pollution and diabetes: A systematic review and meta-analysis

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ABSTRACT

Background: Air pollutants are suggested to be related to type 2 diabetes (T2D). Since several high quality papers on air pollutants and T2D have been published beyond the last reviews, an extended systematic review is highly warranted. We review epidemiological studies to quantify the association between air pollutants and T2D, and to answer if diabetes patients are more vulnerable to air pollutants.

Methods: We systematically reviewed the databases of PubMed and Web of Science based on the guidelines of the Preferred Reporting Items for Systematic review and Meta-analysis (PRISMA). We calculated odds ratios (OR) or hazard ratios (HR) and their 95% confidence intervals (CI) to assess the strength of the associations between air pollutants [e.g., particulate matter with diameter $\leq 2.5 \mu\text{m}$ ($\text{PM}_{2.5}$), particulate matter with diameter $\leq 10 \mu\text{m}$ (PM_{10}), and nitrogen dioxide (NO_2)] and T2D. We evaluated the quality and risk of bias of the included studies and graded the credibility of the pooled evidence using several recommended tools. We also performed sensitivity analysis, meta-regression analysis, and publication bias test.

Results: Out of 716 articles identified, 86 were used for this review and meta-analysis. Meta-analyses showed significant associations of $\text{PM}_{2.5}$ with T2D incidence (11 studies; HR = 1.10; 95% CI = 1.04–1.17 per $10 \mu\text{g}/\text{m}^3$ increment; $I^2 = 74.4\%$) and prevalence (11 studies; OR = 1.08; 95% CI = 1.04–1.12 per $10 \mu\text{g}/\text{m}^3$ increment; $I^2 = 84.3\%$), of PM_{10} with T2D prevalence (6 studies; OR = 1.10; 95% CI = 1.03–1.17 per $10 \mu\text{g}/\text{m}^3$ increment; $I^2 = 89.5\%$) and incidence (6 studies; HR = 1.11; 95% CI = 1.00–1.22 per $\mu\text{g}/\text{m}^3$ increment; $I^2 = 70.6\%$), and of NO_2 with T2D prevalence (11 studies; OR = 1.07; 95% CI = 1.04–1.11 per $10 \mu\text{g}/\text{m}^3$ increment; $I^2 = 91.1\%$). The majority of studies on glucose-homoeostasis markers also showed increased risks with higher air pollutants levels, but the studies were too heterogeneous for meta-analysis. Overall, patients with diabetes might be more vulnerable to PM.

Conclusions: Recent publications strengthened the evidence for adverse effects of ambient air pollutants exposure (especially for PM) on T2D and that diabetic patients might be more vulnerable to air pollutants exposure.

Abbreviations: β , regression coefficient; BC, black carbon; CO, carbon monoxide; CI, confidence interval; GDM, gestational diabetes; GRADE, the Grading of Recommendations Assessment, Development, and Evaluation; HbA1c, glycated haemoglobin; HOMA, homoeostasis model assessment; HOMA-IR, homoeostasis model assessment insulin resistance; HR, hazards ratio; JBI, the Joanna Briggs Institute; NOS, Newcastle-Ottawa Scale; NOx, nitric oxides; O₃, ozone; OHAT, the Office of Health Assessment and Translation; OR, odds ratio; PM, particulate matter; PM_{10} , particulate matter with diameter $< 10 \mu\text{m}$; $\text{PM}_{2.5}$, particulate matter with diameter $< 2.5 \mu\text{m}$; ROB, risk of bias; SO₂, sulfur dioxide; T2D, Type 2 diabetes mellitus; UFP, ultrafine particle

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1. Introduction

Type 2 diabetes mellitus (T2D), characterized by chronic hyperglycemia, is caused by insulin resistance and beta cell dysfunction. T2D has become a leading contributor to the global burden of disease and early death (Forouzanfar et al., 2016; Lim et al., 2018). According to the International Diabetes Federation (Cho et al., 2018), in 2017 approximately 451 million people had diabetes globally, and the prevalence is still increasing, mostly in African, Pacific, and Asian countries (Danaei et al., 2011; Nanditha et al., 2016; NCD risk factors collaboration (NCD Risk Factor Collaboration (NCD-RisC), 2016); van Dieren et al., 2010). In parallel, air pollution has also been listed as one of the major environmental problems worldwide, especially in developing nations (Cohen et al., 2017; Guan et al., 2016). Given the ubiquitous nature of air pollution, it has affected large proportions of people worldwide (Cohen et al., 2017; van Donkelaar et al., 2015). Ambient air pollution (referred to as air pollution hereafter) consists of particulate matter (PM) of different sizes [e.g., $\leq 10\mu\text{m}$ in diameter (PM_{10}); $\leq 2.5\mu\text{m}$ ($\text{PM}_{2.5}$); $\leq 100\text{ nm}$ ultrafine particles (UFP)], and gaseous compounds [e.g., carbon monoxide (CO), nitric oxides (NO_x), nitrogen monoxide (NO), nitrogen dioxide (NO_2), and sulfur dioxide (SO_2), ozone (O_3)]. These compounds show variations in their reactivity, dispersion, and toxicity.

Although accumulating evidence has demonstrated that air pollution adversely affects many health outcomes (Brook et al., 2010; Chen et al., 2013; Cesaroni et al., 2014; Kampa and Castanas, 2008; Pope et al., 2015; Raaschou-Nielsen et al., 2013; Shah et al., 2013; Turner et al., 2011), links between air pollution exposure and T2D have been only recently investigated and the results were still inconsistent. Five recent comprehensive reviews (Dendup et al., 2018; Eze et al., 2015; Janghorbani et al., 2014; Liu et al., 2019; Thiering and Heinrich, 2015) summarized epidemiological studies concerning air pollution and T2D and concluded that evidence indicated a positive association between the two. However, more critical analysis is warranted. First, while most studies detected an increase in the risk of T2D (Eze et al., 2015; Thiering and Heinrich, 2015), other papers published beyond this review failed to detect a significant association (e.g., Coogan et al., 2016a; Coogan et al., 2016b). Additionally, most studies were conducted in North America and Western European countries (Eze et al., 2015; Thiering and Heinrich, 2015). Both the prevalence of T2D and air pollution levels are reported to be much higher in developing countries (Chen et al., 2011; Rohde and Muller, 2015; Yang et al., 2018a), but the effects of air pollution on T2D have not been well studied before 2015 in these regions. For instance, several studies were performed in China in recent two years (Lao et al., 2019; Liang et al., 2019; Yang et al., 2018a, 2018b). The most recent meta-analysis by Liu et al. (2019) only pooled data on long-term air pollution exposure and T2D prevalence or incidence, without including short-term studies and those investigating air pollution and glucose-homoeostasis markers.

To provide healthcare policy makers and researchers with more comprehensive evidence on ambient air pollution and T2D and glucose-homoeostasis markers, we performed an updated systematic review and meta-analysis of studies that investigated relationships of air pollutants with T2D and glucose-homoeostasis markers. We address three fundamental questions: (i) is air pollution exposure associated with increased risks of diabetes development and diabetes-related mortality, (ii) is air pollution associated with markers of glucose homeostasis that predispose to diabetes, and (iii) are participants with diabetes more vulnerable to air pollutants exposure than those without diabetes?

2. Methods

2.1. Search strategy

We conducted the study using the standard methods following the Preferred Reporting Items for Systematic Review and Meta-Analysis

(PRISMA) (Shamseer et al., 2015). Two electronic databases (i.e., PubMed and Web of Science) were searched for peer-reviewed articles published before 25 May 2019 and written in English. The combinations of following terms concerning air pollution (i.e., air pollution, particulate matter, particles, $\text{PM}_{2.5}$, PM_{10} , $\text{PM}_{2.5-10}$, black smoke, black carbon, NO_x , NO_2 , SO_2 , CO, and O_3) and T2D [i.e., diabetes, type 2 diabetes, T2D, gestational diabetes, glucose, insulin resistance, homeostasis model assessment (HOMA), impaired glucose tolerance, and glycated haemoglobin (HbA1c)] were used for the literature search. Additionally, reference lists of all the included studies and previously published systematic reviews (Balti et al., 2014; Dendup et al., 2018; Dimakakou et al., 2018; Eze et al., 2015; He et al., 2017; Janghorbani et al., 2014; Liu et al., 2019; Park and Wang, 2014; Thiering and Heinrich, 2015; Wang et al., 2014) were manually searched to find more studies.

2.2. Study selection and eligibility criteria

Two authors (BYY and SF) performed study selection, and if there was any uncertain then the third author checked (JH). Briefly, we screened potential studies through the following four steps (Fig. S1). First, we downloaded all reference records from the two electronic databases into a reference manager (NoteExpress V3.1, Aegean Software, China). Second, duplicates were deleted from the initial reference records using the software's function. Third, the titles and abstracts of the remaining articles were reviewed for eligibility. Fourth, the full texts of the potentially eligible references were further evaluated.

Studies finally included had to meet the following five criteria: (1) assessed the associations of air pollution with T2D or pre-diabetic conditions, such as fasting blood glucose (FBG), insulin resistance, and impaired glucose tolerance; (2) cohort, case-crossover, time series, cross-sectional, case-control, or panel studies, but not reviews, letters, clinical research studies, or animal experiments; (3) focused on outdoor but not indoor air pollution; (4) written in English; (5) for duplicate publication, we only included the study that reported the most recent and detailed information.

2.3. Data extraction

We extracted the following information from each eligible study: authors' names, publication year, participants' age and gender, exposure assessment of air pollution, exposure period [long-term (≥ 30 days) or short-term (< 30 days)], effect estimates [e.g., odds ratio (OR), and hazards ratio (HR)] and their corresponding 95% confidence intervals (CI), study setting, and sample size. Two independent authors (BYY and SF) extracted this information, and discordances were discussed with a third author (JH) and resolved by consensus.

2.4. Assessment of study quality and risk of bias

Two authors (BYY and SF) independently evaluated the quality and risk of bias (ROB) of the included studies, and any conflicts was adjudicated by a third author (JH). We employed Newcastle-Ottawa Scale (NOS) (Wells et al., 2013) to evaluate the quality of cohort and case-control studies. Each study was assessed on an NOS score from 0 to 9, and the score greater than 7 was classified as "high quality", otherwise the study was categorized as "low quality". We used the scale recommended by Joanna Briggs Institute (JBI) (<http://www.joannabriggs.edu.au/>) to assess the quality of cross-sectional study. Each study can be assessed on a JBI score from 0 to 20, and the score greater than 16 was considered as "high quality". For time-series and case-crossover studies, there is no validated scale to assess their quality. Thus, we adopted a scale adapted from validated NOS scale and the Cochrane risk of bias tool to evaluate these studies (Mustafic et al., 2012). The adapted scale included validation of T2D or its traits (0–1 point), the quality of air pollution assessment (0–1 point), and the

extent of covariates adjustments (0–3 points). If the maximum score was achieved for all the three items (i.e., 5 points), the study was classified as “high quality”, otherwise “low quality”.

We further employed the Office of Health Assessment and Translation (OHAT, 2015) to assess the ROB for cohort and cross-sectional studies. As there is no validated tool for assessing ROB of time series and case-crossover studies, we thus employed the tool recommended by the National Institute of Environmental Health Sciences National Toxicology Program (OHAT, 2015) and the Navigation Guide by the University of California for these studies (Lam et al., 2016). The included studies were assessed for key criteria (i.e., assessment of exposure and outcome, and confounding bias) and other criteria (i.e., attrition/exclusion bias, selection bias, reporting bias, conflict of interest, and other bias). Each of the criteria was assessed as “low”, “probably low”, “probably high”, or “high” risk. If a study for which the key criteria and most of the other criteria are categorized as “high” or “probably high”, then the study should be removed from the review.

2.5. Statistical analysis

Due to apparent differences in characteristics (e.g., study design, participants' health status, exposure assessment, and outcomes) for studies investigating diabetic mortality in general population, all-cause mortality in diabetic patients, and other diabetic traits (e.g., FBG, insulin, insulin resistance), we only performed meta-analyses for air pollutants with T2D prevalence or incidence. In addition, we only focused on PM and NO₂, as the number of studies on other air pollutants was limited.

For air pollutants with T2D prevalence or incidence available from more than four studies, we extracted point effect estimates (OR or HR with their corresponding 95% CI) from individual studies and then pooled them using the inverse variance method. The Z test was used to test the significance of the pooled estimates. The between-study heterogeneity was evaluated using both I² statistics and Q-test (Higgins and Thompson, 2011). If I² ≥ 50% or P value for Q-test was < 0.01, then the heterogeneity was considered as “high” and the Der Simonian and Laird random effects model was used to pool effect estimates. Otherwise, the heterogeneity was considered as “moderate or low” and the Mantel-Haenszel fixed effects model was used to pool effect estimates (Der Simonian and Laird, 1986). For meta-analysis with 10 data sets or more (Higgins and Green, 2011) and with high between-study heterogeneity, we performed univariate meta-regression to explore potential sources of the heterogeneity. The covariates included publication year, study settings, study design, exposure assessment, sample size, participants' age and sex, background air pollutants levels, and diagnosed method of T2D. We also performed sensitivity analyses by excluding each study. Egger's regression test was used to evaluate potential publication bias (Egger et al., 1997).

As only a small part of studies applied multi-pollutant model, we extracted and pooled effect estimates from single-pollutant model. In addition, most published studies used a generalized linear model, thus we assumed a linear association and pooled all effects estimates for a fixed 10 µg/m³ increase in air pollutant concentration. Other reported units or scales were harmonized as published elsewhere (Yang et al., 2018c).

Finally, we employed the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) Working Group to grade the credibility of the pooled evidence.

All the above statistical analyses were performed using STATA package version 12.1 program (StatCorp, College Station, TX, USA). A P-value < 0.05 was considered as statistically significant.

3. Results

3.1. Literature retrieval and study characteristics

The systematic literature searches yielded a total of 716 hits (Fig. S1). After excluding duplicates and other publications according to the eligibility assessment by abstract and full text screens, 86 articles were identified as eligible for this review. The basic characteristics of the 86 selected publications are shown in Tables S1–S3 of the supplement, which were grouped by study design and outcomes [diabetes mortality, diabetes incidence and prevalence, diabetes-related traits, and gestational diabetes (GDM)], and were ordered by date of publication. The included studies were from 19 countries, most of which are western developed nations such as North America and European countries. Cohort and cross-sectional are the most commonly used study designs. About two thirds of the included studies assessed air pollution exposure using models (e.g., land use regression model, hybrid spatial-temporal model, satellite-based model), and the remaining studies used data from fixed air monitoring stations. Twenty studies adopted multi-pollutant models, but the pattern of the associations were mixed; while some studies found that the effect estimates in the single-pollutant models were attenuated by additionally adjusted for other pollutant(s), the others observed that the effects were strengthened or not materially affected (Tables S1–S3). Concerning the study quality assessment, 57 studies were considered as having “high quality” and the remaining studies were evaluated as “low quality” (Table S4). With respect to the ROB assessment, none of the included studies was excluded because of high ROB (Table S4).

3.2. Is exposure to air pollution associated with an increased risk of diabetes?

To answer this question, we will make use of study findings on mortality caused by diabetes in the general population as well as incidence and prevalence of T2D in prospective cohorts and cross-sectional studies. Detailed information of the included studies is summarized in Tables S1–S3 of the supplement.

3.2.1. Effects of air pollution on diabetes-related mortality in the general population

Eleven population-based studies have investigated the relationships for short-term exposure to air pollution and diabetes-related mortality (Fig. 1 and Table S1). All studies, except one (Zuniga et al., 2016), suggested an increase in daily diabetes-caused mortality after exposure to days of high PM or black carbon (BC), and six out of the 11 studies reported statistically significant increased mortality risks (e.g., a 10-µg/m³ increment in PM_{2.5} was associated with 6–49% increase in diabetes-related mortality). Due to high heterogeneity in design (time series, case-crossover) and the limited number of such studies, a meta-analysis was not performed.

Four cohort studies explored long-term exposure to air pollution and T2D related mortality. Higher levels of PM_{2.5} and NO₂ were associated with greater T2D mortality in all the four studies, but no significant association was observed for O₃ (Table S1).

3.2.2. Long-term effects of air pollution on T2D incidence

Prospective studies follow population at risk (i.e., participants without diabetes) for a certain time and investigated relationships of air pollution exposure at baseline or during the follow-up period (annual averages levels of air pollutants were often used as surrogates of long-term exposure) with the incidence of T2D cases. Fig. 2 illustrates the summary findings of overall increased incidence of T2D with increased

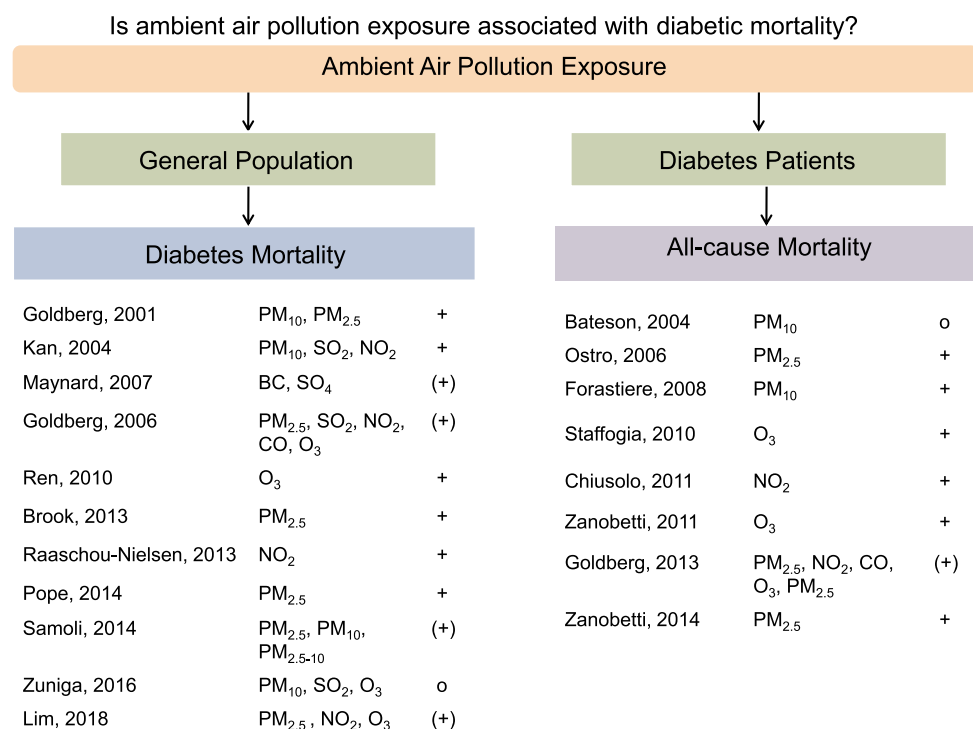


Fig. 1. Overview of studies examining diabetes mortality in the general population and all-cause mortality in diabetes patients. The direction of the effect is denoted by “+” for positive association, “o” for no association, and “(+)” for non-significant associations or associations that are restricted to subgroups. For short-term exposure studies that reported multiple lags, we showed the results from the lag that the authors stated as a priority, the lag was statistically significant, or the lag that reported the largest effect estimates. PM₁₀ indicates particulate matter with diameter < 10 µm; PM_{2.5}, particulate matter with diameter < 2.5 µm; NO₂, nitrogen dioxide; NOx, nitric oxides; O₃, ozone; BC, black carbon; CO, carbon monoxide; SO₂, sulfur dioxide; SO₄, sulfate.

exposure to at least one air pollutant. Fifteen out of 21 studies reported statistically significant increased risks with air pollutants.

In meta-analysis, the pooled effect for T2D incidence was 1.10 (95% CI 1.04–1.17) and 1.11 (95% CI 1.00–1.22) for an increment of 10 µg/m³ in PM_{2.5} and PM₁₀, respectively, based on 11 and six studies (Fig. 3 and Table S5). We also meta-analyzed seven studies investigating the risk for the development of T2D in relation to NO₂ and observed that pooled effect was 1.01 (95% CI 0.99–1.02) (Fig. 3 and Table S5). High between-study heterogeneity was observed for all the above three meta-analyses (Table S5). Meta-regression analysis for PM_{2.5} and DM incidence showed that none of the investigated covariates was a substantial source of heterogeneity (data not shown). Sensitivity analyses by excluding each study did not change the overall estimates (Figs. S2–S4). Publication bias was detected for the meta-analyses for PM_{2.5} and PM₁₀ with T2D incidence (Table S5; Figs. S5–S7). Based on the GRADE system, confidence in the pooled evidence was “very low” for both PM_{2.5} and PM₁₀, and moderate for NO₂ (Table S5).

Two prospective studies investigated ozone and T2D, and both reported increased risks (Jerrett et al., 2017; Renzi et al., 2018) (Fig. 2).

3.2.3. Long-term effects of air pollution on T2D prevalence

A total of 25 cross-sectional studies were performed to evaluate relationships for air pollution and T2D prevalence and results are illustrated in Fig. 2 and Table S2. The vast majority of these studies (18 out of 25) showed increased risks for T2D prevalence for at least one of the investigated pollutants. For meta-analysis, we extracted 11 studies on PM_{2.5}, six on PM₁₀, and 11 on NO₂ with T2D prevalence and found increased risk (Fig. 4 and Table S6). The overall effect estimate was increased by 1.08 (95% CI 1.04–1.12), 1.10 (95% CI 1.03–1.17), and 1.07 (95% CI 1.04–1.11) for an increment of 10 µg/m³ of PM_{2.5}, PM₁₀, and NO₂, respectively (Fig. 4 and Table S6). Sensitivity analyses by excluding each study did not change the overall estimates (Figs. S8–S10), and evidence of publication bias was detected for all the three air pollutants (Table S6; Figs. S11–S13). We detected high between-study heterogeneity for all the three meta-analyses (Table S6). Meta-regression analysis showed that none of the investigated covariates (e.g. study setting, exposure assessment, sample size, age, and diagnosed

method of T2D) was the potential source of the heterogeneity (data now shown). Based on the GRADE system, the credibility of all the pooled evidence was “low” or “very low” (Table S6).

3.3. Is exposure to air pollution associated with markers of glucose homeostasis?

The results of studies on air pollution and markers of glucose homeostasis are shown in Fig. 5 (for study details see Table S3 of the supplement). Out of 17 studies on fasting blood glucose, 14 reported increased risk for at least one of the investigated pollutants. For example, 14 studies investigated PM_{2.5} and reported that a 10 µg/m³ increment in PM_{2.5} was significantly associated with 0.05–24.7 mg/dL higher levels of fasting blood glucose. Also, HbA1c showed increased risk in seven out of eight studies. Consistent with these findings, homeostatic model assessment of insulin resistance (HOMA-IR) was increased with increased exposure to air pollution in eight out of 10 studies. The results of few studies on 2h-blood glucose and insulin levels are mixed. Evidence has shown that women experiencing gestational diabetes are more likely to have subsequent T2D (Kwak et al., 2013). To date, six out of eight studies reported positive associations between gestational diabetes and exposure to air pollution. For example, in a cross-sectional analysis of 410,267 American women, Hu et al. (2015) found that the risk of gestational diabetes significantly increased 20% per 5 µg/m³ increase in PM_{2.5} (Table S3; Fig. 5).

3.4. Are subjects with diabetes more vulnerable to air pollution exposure than subjects without diabetes?

Case-crossover and time-series studies were performed to compare the effects of short-term air pollution exposure on all-cause mortality among T2D patients and participants without diabetes. As shown in Table S1 and Fig. 1, although not all of the estimates were statistically significant, the cumulative evidence seems to indicate a positive association between short-term air pollution exposure and the daily all-cause mortality risk in diabetic subjects. In addition, most of these studies reported a greater PM-associated mortality risk in diabetic

Is ambient air pollution exposure associated with diabetic prevalence and incidence?

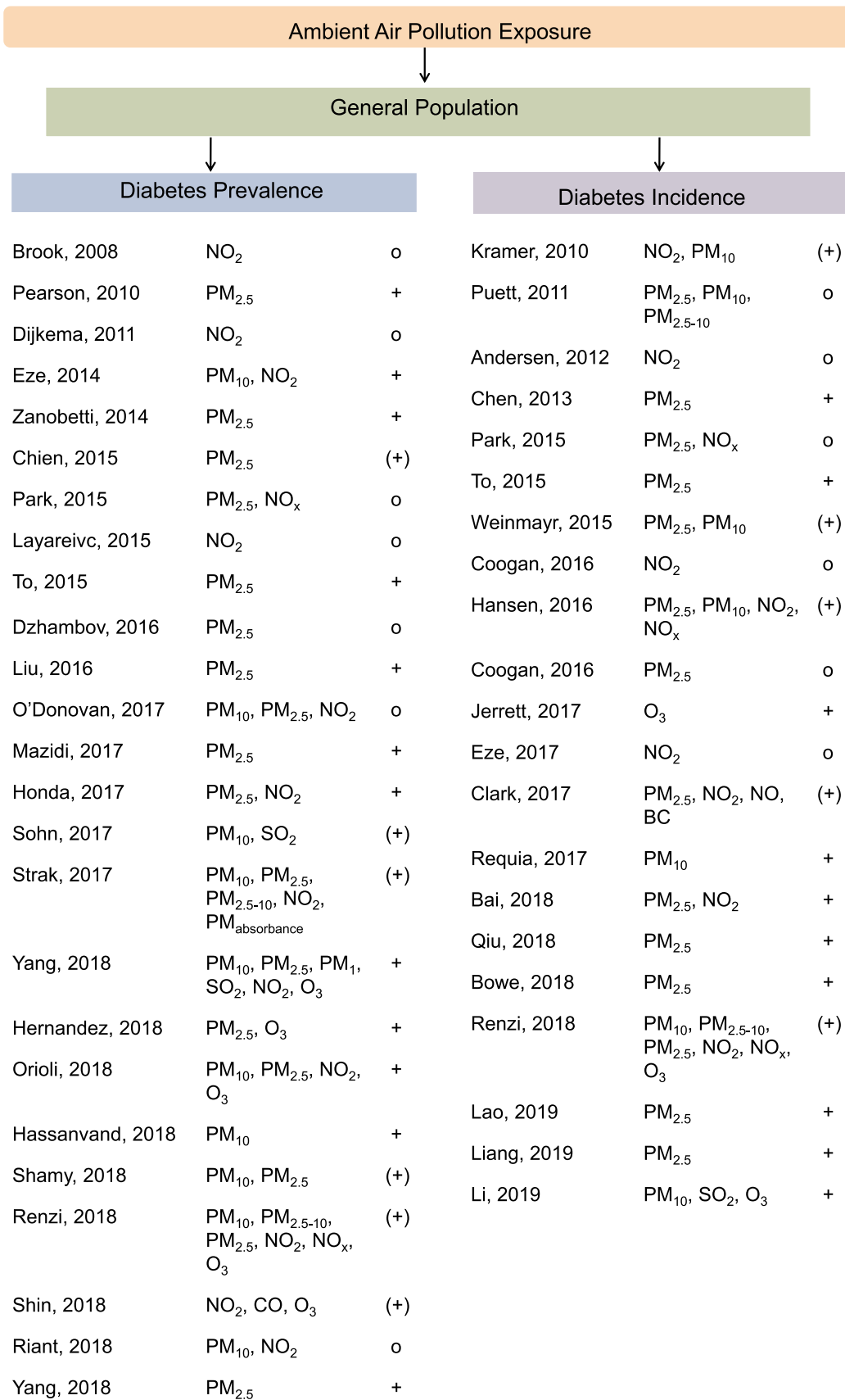


Fig. 2. Overview of studies examining ambient air pollution exposure and its association with diabetes development. The studies investigate associations between ambient air pollution exposure with diabetes incidence and prevalence. The direction of the effect is denoted by “+” for positive association, “o” for no association, and “(+)” for non-significant associations or associations that are restricted to subgroups. PM₁₀ indicates particulate matter with diameter < 10 μm; PM_{2.5}, particulate matter with diameter < 2.5 μm; PM_{2.5-10}, particulate matter with diameter ranging from 2.5 to 10 μm; PM₁, particulate matter with diameter < 1.0 μm; NO₂, nitrogen dioxide; NO_x, nitric oxides; O₃, ozone; BC, black carbon; CO, carbon monoxide; SO₂, sulfur dioxide.

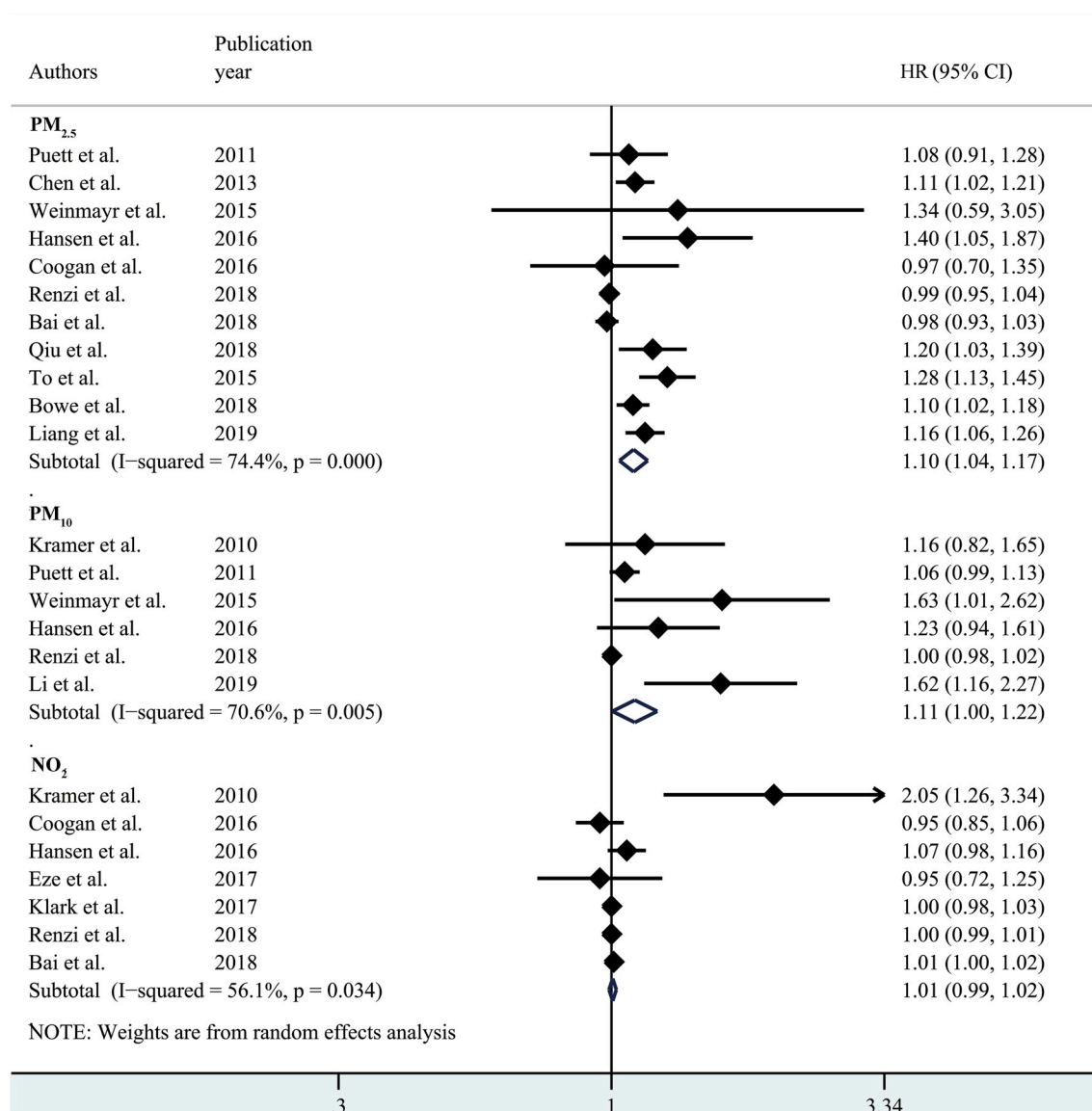


Fig. 3. Meta-analysis of type 2 diabetes incidence and exposure to air pollutants for increments of $10 \mu\text{g}/\text{m}^3$. PM₁₀ indicates particulate matter with diameter < 10 μm ; PM_{2.5}, particulate matter with diameter < 2.5 μm ; NO₂, nitrogen dioxide.

patients than in non-diabetic population (Table S1).

For example, two studies investigated the associations between O₃ exposure and all-cause mortality among diabetes patients (Zanobetti and Schwartz, 2011; Stafoggia et al., 2010), and both reported significant and positive associations (Table S1; Fig. 1). Stafoggia et al. (2010) also observed that the association of O₃ with mortality was stronger in diabetic participants (5.5% (95% CI = 1.4–9.8%) per $10 \mu\text{g}/\text{m}^3$ increment) than in general population (1.5% (95% CI = 0.9–2.1%) per $10 \mu\text{g}/\text{m}^3$ increment).

A multi-center study on the effects of short-term NO₂ exposure reported an increased risk of all-cause mortality risk in diabetic patients [3.61% (95% CI = 1.73–5.53%) per $10 \mu\text{g}/\text{m}^3$ increment], which was greater than those observed in participants without diabetes [2.03% (95% CI = 0.87–3.21%) per $10 \mu\text{g}/\text{m}^3$ increment] (Chiusolo et al., 2011) (Table S1; Fig. 1).

In summary, compared to non-diabetic subjects, the observed up to 2-fold increased risk of mortality suggest that diabetic patients might be more vulnerable to the adverse effects of short-term air pollution exposure (Table S1).

4. Discussion

4.1. Key findings

In the current systematic review and meta-analysis of 86 epidemiological studies conducted in 19 countries, we performed a comprehensive evaluation of available data on ambient air pollution and T2D. Most short-term exposure studies reported a positive association between higher air pollution levels and greater diabetes-caused mortality. Meta-analytical results showed significant associations of PM_{2.5}, PM₁₀, and NO₂ with T2D prevalence, and of PM_{2.5} and PM₁₀ with T2D incidence. However, the between-study heterogeneity for these meta-analyses was high, which significantly reduced the credibility of the pooled evidence. Significant association was also found for air pollutants and glucose homeostasis markers (especially for fasting blood glucose, HbA1c, and HOMA-IR) and gestational diabetes, but the studies were varied in study design, outcome definition, exposure assessment, and the number of studies for some pollutants were limited. Further, several short-term studies indicated higher mortality rates among diabetic patients than those in non-diabetic individuals.

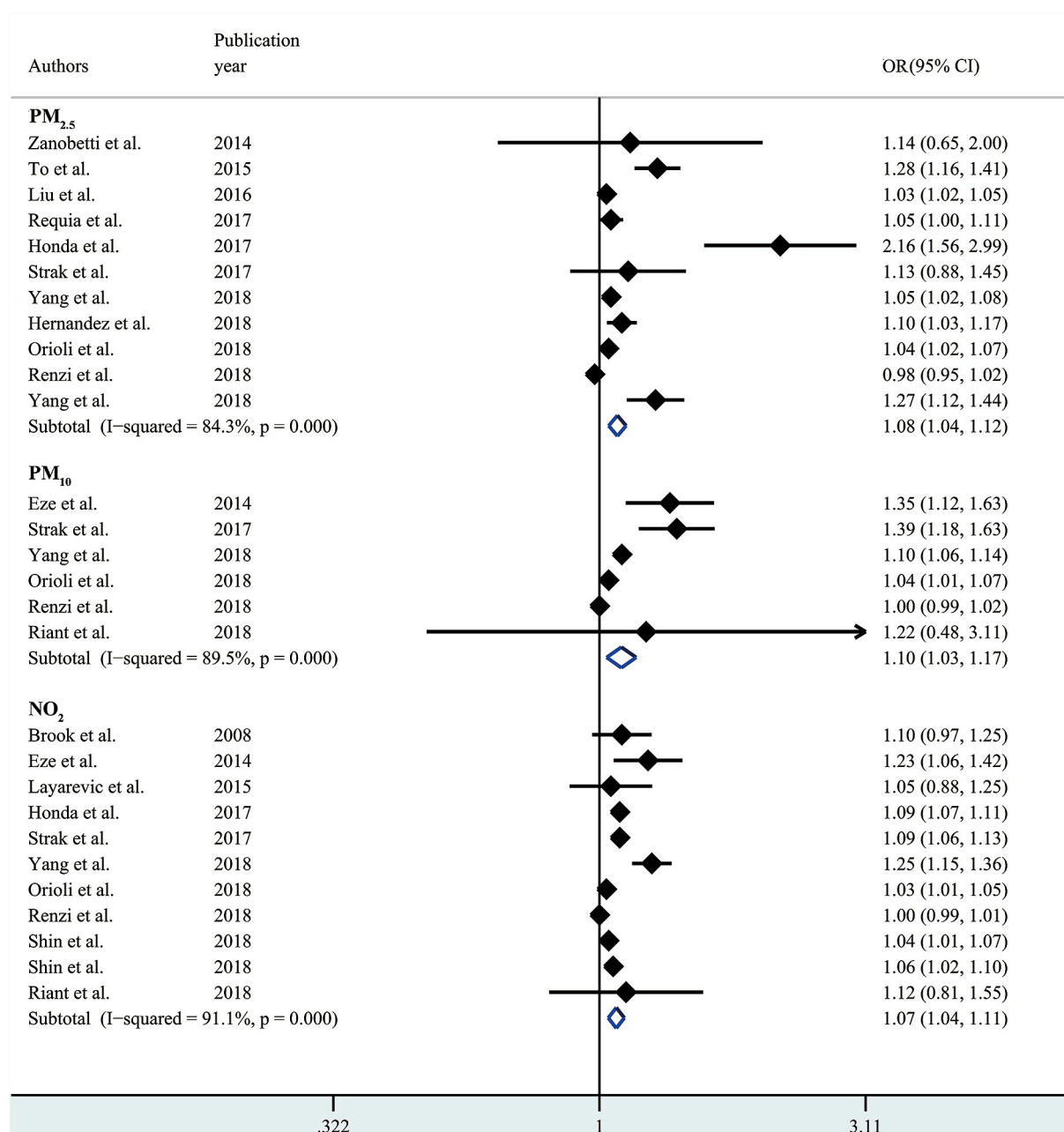


Fig. 4. Meta-analysis of type 2 diabetes prevalence and exposure to air pollutants for increments of $10 \mu\text{g}/\text{m}^3$. PM₁₀ indicates particulate matter with diameter < 10 μm ; PM_{2.5}, particulate matter with diameter < 2.5 μm ; NO₂, nitrogen dioxide.

4.2. Comparison with prior meta-analyses and interpretations

Studies for air pollutants with T2D morbidity and mortality published before 2015 have been partly analyzed in five recent meta-analyses (Balti et al., 2014; Eze et al., 2015; Janghorbani et al., 2014; Park et al., 2014; Wang et al., 2014). All of these published meta-analyses documented a positive association for long-term exposure to air pollutants and T2D, but the included studies, the methodology, and the results of these reviews and meta-analyses are different (Thiering and Heinrich, 2015). In addition, the conducted studies have limitations in correlation with other unmeasured air pollutants, uncontrolled confounding (e.g., physical activity, smoking), other environmental exposures (e.g., indoor, occupational, smoking, greenspace), and comorbidities (Eze et al., 2014). A few more recent systematic review articles also reported statistically significant associations between air pollutants and T2D, but were incomplete or did not perform meta-

analysis (Dimakakou et al., 2018; He et al., 2017). More specifically, the systematic review by Dimakakou et al. (2018) only included 18 studies investigating air pollution and T2D and diabetes traits. The meta-analysis by He et al. (2017) included 11 cohort studies on PM_{2.5} and T2D incidence and reported that a $10 \mu\text{g}/\text{m}^3$ increase in PM_{2.5} was associated with a 25% higher risk of T2D. The most recent meta-analysis by Liu et al. (2019) meta-analyzed 30 studies published before 2018, and reported significant associations for PM₁₀, PM_{2.5}, and NO₂ with T2D prevalence, and for PM_{2.5} with T2D incidence. By comparison, the results from these published meta-analyses were not exactly the same as those from our current study, which might be caused by the different number of the included studies. We updated the literature search up to May 25, 2019, which generated more studies and participants (Table S7). Thus, we believe our estimates would be more comprehensive. Despite this, it is notable that the between-study heterogeneity in our meta-analyses was high, which substantially reduced the credibility of

Diabetes Related Traits and GDM prevalence

FBG			Yang, 2018	PM ₁ , PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , O ₃	(+)
Chuang, 2011	PM ₁₀ , PM _{2.5} , NO ₂ , O ₃	(+)			
Brook, 2013	PM _{2.5}	+	HOMA-IR		
Kim, 2014	PM ₁₀ , NO ₂ , O ₃	+	Kelishadi, 2009	PM ₁₀ , CO	+
Sade, 2015	SO ₂ , NO ₂	(+)	Brook, 2013	PM _{2.5}	+
Chen, 2016(a)	PM ₁₀ , NO ₂ , SO ₂	+	Chuang, 2011	PM ₁₀ , PM _{2.5} , O ₃ , NO ₂ , SO ₂	(+)
Chen , 2016(b)	PM _{2.5} , NO ₂	+	Kim, 2014	PM ₁₀ , NO ₂ , O ₃	(+)
Yitshak Sade, 2016	PM ₁₀ , PM _{2.5}	(+)	Chen , 2016(b)	PM _{2.5} , NO ₂	+
Liu, 2016	PM _{2.5}	+	Thiering, 2016	PM ₁₀ , PM _{2.5} , PM _{absorbance} , NO ₂	(+)
Lu, 2017	PM _{2.5}	+	Yang, 2018	PM ₁ , PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , O ₃	o
Alderete, 2017	PM _{2.5} , NO ₂	o	Li, 2018	PM _{2.5}	o
Lucht, 2018	PM _{2.5} , PM ₁₀ , NO ₂	(+)	Toledo-Corral, 2018	PM _{2.5} , NO ₂ , NO _x , O ₃	(+)
Yang, 2018	PM ₁ , PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , O ₃	+	Khafaie, 2018	PM ₁₀	+
Li, 2018	PM _{2.5}	o	HbA1c		
Toledo-Corral, 2018	PM _{2.5} , NO ₂ , NO _x , O ₃	(+)	Chuang, 2011	PM ₁₀ , PM _{2.5} , NO ₂ , O ₃	(+)
Khafaie, 2018	PM ₁₀	o	Yitshak Sade, 2016	PM ₁₀ , PM _{2.5}	+
Erqou, 2018	PM _{2.5}	+	Liu, 2016	PM _{2.5}	+
Riant, 2018	PM ₁₀ , NO ₂	(+)	Honda, 2017	PM _{2.5} , NO ₂	+
Fasting insulin			Lucht, 2018	PM _{2.5} , PM ₁₀ , NO ₂	(+)
Brook, 2013	PM _{2.5}	+	Li, 2018	PM _{2.5}	o
Kim, 2014	PM ₁₀ , NO ₂ , O ₃	(+)	Khafaie, 2018	PM ₁₀	+
Chen , 2016(b)	PM _{2.5} , NO ₂	(+)	Riant, 2018	PM ₁₀ , NO ₂	+
Alderete, 2017	PM _{2.5} , NO ₂	(+)	GDM		
Yang, 2018	PM ₁ , PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , O ₃	o	Malmqvist, 2013	NO _x	+
Toledo-Corral, 2018	PM _{2.5} , NO ₂ , NO _x , O ₃	(+)	Fleisch, 2014	PM _{2.5} , BC	o
2h-BG			Hu, 2015	PM _{2.5} , O ₃	+
Chen , 2016(b)	PM _{2.5} , NO ₂	o	Robledo, 2015	PM ₁₀ , PM _{2.5} , SO ₂ , O ₃ , NO _x	(+)
Lu, 2017	PM _{2.5}	o	Fleisch, 2016	PM _{2.5}	o
Alderete, 2017	PM _{2.5} , NO ₂	o	Pedersen, 2017	NO ₂	+
Yang, 2018	PM ₁ , PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , O ₃	(+)	Shen, 2017	PM _{2.5} , SO ₂ , NO ₂ , O ₃	(+)
Khafaie, 2018	PM ₁₀	+	Pan, 2017	PM ₁₀ , CO, NO, NO _x , SO ₂ , O ₃	(+)
2h-insulin					
Chen , 2016(b)	PM _{2.5} , NO ₂	o			
Alderete, 2017	PM _{2.5} , NO ₂	(+)			

Fig. 5. Overview of studies examining ambient air pollution exposure and its association with predisposing conditions for diabetes. The studies on ambient air pollution exposure in relation to fasting blood glucose (FBG), fasting insulin, 2h-blood glucose (2h-BG) and insulin (2h-insulin), glycated haemoglobin (HbA1c), homeostatic model assessment of insulin resistance (HOMA-IR), and gestational diabetes (GDM). The direction of the effect is denoted by “+” for positive association, “o” for no association, and “(+)” for non-significant associations or associations that are restricted to subgroups. For short-term exposure studies that reported multiple lags, we showed the results from the lag that the authors stated as a priority, the lag was statistically significant, or the lag that reported the largest effect estimates. PM_{10} indicates particulate matter with diameter $< 10 \mu m$; $PM_{2.5}$, particulate matter with diameter $< 2.5 \mu m$; $PM_{2.5-10}$, particulate matter with diameter ranging from 2.5 to $10 \mu m$; PM_1 , particulate matter with diameter $< 1.0 \mu m$; NO_2 , nitrogen dioxide; NOx , nitric oxides; O_3 , ozone; BC, black carbon; CO, carbon monoxide; SO_2 , sulfur dioxide.

the accumulative evidence. We incorporated a set of covariates into meta-regression analysis to detect potential sources of the heterogeneity, but found that none of the study-level variables was responsible for the heterogeneity. Therefore, other unmeasured or unreported variables would be more important in explaining the heterogeneity.

For studies investigating short-term effects of air pollution on diabetic caused mortality, a cautious interpretation is needed, although half of the effect estimates are significantly increased. These studies cannot disentangle between short-term triggering effects of PM and long-term effects, because it is mostly not known whether exposed subjects developed T2D and further exposure is just triggering mortality with underlying diabetes. However, evidence from prospective cohort studies has indicated a positive association between long-term exposure to air pollution and T2D mortality, albeit the number of such cohort studies was small.

Studies on markers of glucose homeostasis, which might be changed in subjects with a higher risk of diabetes, can help shed light on the association of air pollution exposure with diabetes and the potential biological mechanisms underlying the association. Two very recent systematic reviews and meta-analyses (Dang et al., 2018; Elshahidi, 2019) have pooled the results on air pollution with HOMA-IR and gestational diabetes by including 6 and 8 studies, respectively. Dang et al. (2018) observed that long-term exposure to NO₂ and PM₁₀, but not PM_{2.5}, was associated with higher risk of insulin resistance. However, they found serious publication bias for the included studies, which reduced the credibility of the pooled evidence. Elshahidi (2019) investigated the associations between seven air pollutants (CO, NO, NO₂, NO_x, O₃, SO₂, PM₁₀, and PM_{2.5}) and gestational diabetes, and found significant associations for NO and SO₂. Likewise, the included studies in that meta-analysis have high heterogeneity in study design, exposure assessment, and outcome definition, and the number of studies for each air pollutant was limited, which restricts the ability to obtain a more precise estimate with minimal heterogeneity. Thus, more well-designed longitudinal studies remain warranted to explore these topics.

4.3. Potential biological mechanisms

Although the accurate pathways underlying air pollution and diabetes remain unclear, several mechanisms have been proposed. Evidence has well-demonstrated that PM exposure is associated with elevated systemic inflammation and oxidative stress, endoplasmic reticulum stress, impaired endothelial function, cardiac autonomic nervous system dysfunction, and mitochondrial dysfunction (Brook et al., 2010; Rajagopalan and Brook, 2012; Rückerl et al., 2014; Liu et al., 2013, 2016). Additionally, evidence also showed that exposure to air pollutants can cause epigenetic changes, which may lead to changes in markers of coagulation, inflammation, and endothelial function (Bind et al., 2012; Carmona et al., 2014). Moreover, a few animal studies suggested a possible role of PM_{2.5} exposure and insulin resistance (Sun et al., 2009; Liu et al., 2014; Xu et al., 2011). Finally, the effect modification of air pollution and T2D by a genetic risk score provided an additional argument for a causal interrelationship between air pollution and T2D (Eze et al., 2016).

4.4. Strengths and limitations

A major strength of our review is that we provided the most systematic assessment of the current epidemiological evidence on ambient air pollution and T2D. We covered many air pollutants (e.g., PM_{2.5}, PM₁₀, NO₂, SO₂, O₃, CO) and a rich set of T2D metrics (e.g., prevalence, incidence, mortality, FBG, HOMA-IR, HbA1C), which otherwise would be difficult to be obtained from individual studies or isolated reviews or meta-analyses. In addition, we assessed the quality and ROB of each study and graded the credibility of the pooled evidence, which might be useful for researchers and policy makers to understand the limitations

and gaps in the existing evidence.

However, some limitations should be acknowledged. First, the included studies differed markedly in outcome assessment (e.g., self-reported versus diagnosed diabetes), exposure assessment (e.g., measurements from air-monitoring stations versus spatial-statistical model), and exposure period (e.g., lag days for short-term exposure and exposure window of long-term exposure), which might have biased effect estimates. In meta-analysis, we also detected high between-study heterogeneity, which reduced the credibility of the cumulative evidence, and thus healthcare policy makers and researcher should be cautious in using and interpreting the evidence. We also performed meta-regression by considering a set of covariates, but did not detect any major source of heterogeneity. The unmeasured or unreported variables such as population sensitivity and other potential sources and constituents of air pollutants might be more responsible for the heterogeneity. Second, we employed the random effects model to pool these data; however, the method also has limitations in overestimating effects and underestimating the statistical error (Doi et al., 2015). Third, although several recent studies have been performed in developing countries such as China, most of the included studies were from developed countries. Thus, the generalizability of the current evidence to other highly polluted areas is limited, especially considering possible non-linear relationship between air pollution and health outcomes (Burnett et al., 2014). Fourth, while most studies focused on PM and NO₂, the diabetic effects of other air pollutants (e.g., O₃) were less studied. Fifth, one third of the included studies used air pollutant monitoring data but not personal exposure measurements. In addition, while some studies objectively diagnosed diabetes, the others defined diabetes using a self-report of doctor diagnosis or registered data. These assessments might have misclassified actual exposure and outcomes and thus underestimated the effect estimates. Sixth, the number of covariates differed among studies and several important covariates (e.g., noise and greenspace) were not controlled in most studies. Thus, residual confounding is still possible. Seventh, due to high correlation between air pollutants, investigators usually adopted single-pollutant model to generate effect estimates, while it is likely that the diabetic effects of one pollutant is masked or dominated by the other pollutant(s). Eighth, we mainly focused on the studies conducted in adults and did not explore the health effects of air pollution on T2D during different stages of the life course as Lim and Thurston (2019) published recently.

4.5. Concluding remarks and future perspectives

The evidence we reviewed indicates an association of long-term exposure to air pollution with T2D development and diabetes-related mortality in adults. The results from studies assessing biomarkers of glucose homeostasis in healthy individuals provide more support for a potential association. However, because of limited studies available, a robust conclusion on the association of air pollution exposure and diabetes-related traits and GDM cannot be presently drawn. For short (er)-term air pollution effects in diabetic patients, the recent overall results strengthened the evidence for an adverse triggering effect of air pollution on acute mortality. However, these effects can be partly due to long-term exposure and T2D development, especially considering that those short-term studies did not adjust for long-term air pollution exposure. Therefore, results of prospective cohort studies are more valid for the assessment of air pollution effects on mortality cause by T2D.

In conclusion, overall and in particular the recently published studies strengthened substantially the evidence for adverse effects of air pollution exposure on T2D. Future studies are also needed to clarify the role of air pollution exposure at other places than outdoors and T2D and to investigate how and to what extent air pollution control strategies can reduce the burden of diabetes-related diseases.

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Ethics statement

The study was conducted with the principles stipulated by the Declaration of Helsinki and all procedures were approved by the ethics review committee of Sun Yat-Sen University.

Declaration of competing interest

We declare no competing interests.

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Appendix A. Supplementary data

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